Encirra

A Coupled Power–Water Digital Twin for Verified Recovery from Cascading Urban Infrastructure Failures

See more. Recover faster. Safer cities for tomorrow.

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NEECON 2026 Undergraduate Innovation Challenge
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17:20 · Feeder F3 fault, Encirra recovery applied. Red: isolated faulted section. Green: cables re-energised by the verified plan. Amber: healthy supply. Cyan: water mains in the exposed underground. Labels and colours are driven by the power-flow and hydraulic solutions.

1 The problem: failures cross domains

Urban water networks depend on electrically driven pumps. When a distribution feeder trips, the **power** outage is visible immediately, but the **water** consequence is hidden: elevated tanks keep pressure up for tens of minutes, then critical facilities such as hospitals lose supply.

Conventional fault restoration maximises restored electrical load. It treats a pump as an ordinary load and counts a hospital on its standby generator as "served", so it can choose a plan that is electrically optimal and still lets the hospital run out of water.

72 min
from pump trip to hospital pressure < 20 m (simulated)

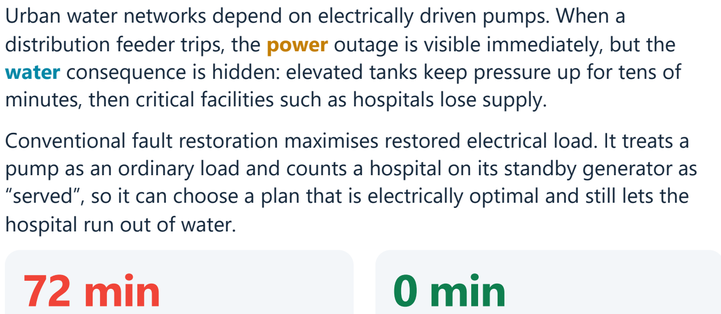
0 min
below service pressure with Encirra's verified plan

2 What Encirra is

A physics-driven, optimization-assisted digital twin that **models** the coupled networks, **predicts** the cascade, **evaluates** recovery plans across both domains and **independently verifies** them before they are recommended.

- **Electrical:** IEEE 33-bus distribution feeder (Baran & Wu) solved with AC power flow in pandapower; 37 switchable lines, 5 normally-open ties.
- **Hydraulic:** two-zone network (2 pumping stations, 2 elevated tanks, desalination supply) solved as a pressure-driven extended-period simulation with EPANET 2.2 via WNTR.
- **3D twin:** a Dubai-inspired district with an exposed underground; every building is mapped to the bus that supplies it.

3 Coupling the two models



$$P_{pump} = \rho \cdot g \cdot Q \cdot H / \eta$$

electrical demand from the hydraulic solution (W → kW), fed back into the AC power flow

If a pump's bus is de-energised or below 0.90 pu, the pump stops and the hydraulic model is re-solved, so flow, pressure and tank levels always come from the recalculated simulation, never from a scripted curve.

4 Recovery planning

After protection isolates the fault, Encirra generates radial reconfigurations from remote-operable switches (closing up to two ties, opening a sectionaliser, optionally shedding non-critical load) and simulates every candidate on the coupled model over the remaining horizon.

Strategy	Decision rule
None	Baseline: the outage persists.
Grid-only	Conventional restoration: maximise restored electrical energy within hard limits.
Encirra	Weighted objective over served load, critical load, water delivered, critical-facility water service, pressure/storage margin and switching cost.

Weights are visible and configurable. Hard safety constraints are never traded off against the score: a candidate that fails any of them is rejected outright.

5 Independent verification

The selected plan is re-simulated minute by minute on networks rebuilt through a separate code path. Only plans that pass every hard check are labelled **Simulation Verified**.

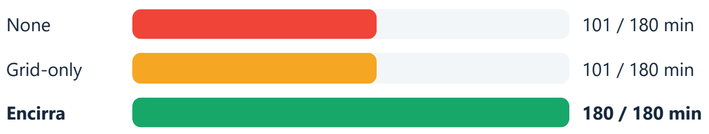
- ✓ Power-flow convergence
- ✓ Thermal line ratings
- ✓ Fault stays isolated
- ✓ Critical pressure ≥ 20 m
- ✓ Remote switching only
- ✓ Voltage band 0.90–1.05 pu
- ✓ Radiality & connectivity
- ✓ Pump supply available
- ✓ Tank above minimum
- ✓ Critical facility service

Feeder-fault scenario: **80 candidates** evaluated, **59 rejected** with their physical reason, e.g. **V(B29) = 0.775 pu < 0.90 pu** or **line B25–B29 loaded 130 % > 100 %**.

6 Results

Scenario	Strategy	Net restored	PS-2	Hospital min. p	Min < 20 m
Feeder fault	None	0 %	off	0.0 m	79
	Grid-only	69.2 %	off	0.0 m	79
	Encirra	**78.7 %**	**on**	**37.8 m**	**0**
Pump power loss	Grid-only	91.7 %	off	0.0 m	79
	Encirra	91.2 %	**on**	**37.8 m**	**0**
High demand	Both (same plan)	78.6 %	on	37.1 m	0

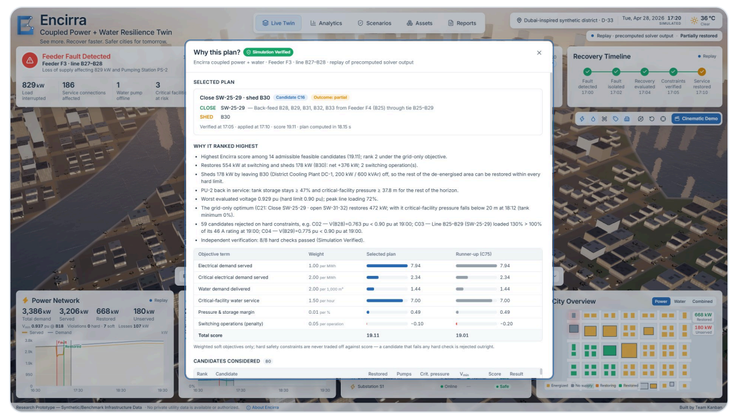
Feeder fault: hospital water service over the 3-hour horizon



Every selected plan passed every hard check. Under high demand both planners converge on the same plan, and Encirra's advantage depends on the synthetic tie-line capacity; both findings are reported with a sensitivity analysis.

7 The cascade, side by side

Grid-only at 18:20. Electrically verified, yet PS-2 stays dark: Tank T2 has emptied and hospital pressure is 0 m, 72 minutes after the trip.



"Why this plan?" Every candidate, its score breakdown, rejected alternatives with their failed checks, and the verification result.

8 Contribution & next steps

- An open, reproducible coupled power–water co-simulation with explicit pump-power feedback.
- Cross-domain recovery that protects critical water service, with independent verification and transparent evidence for every decision.
- Validated on a published benchmark (Vmin 0.913 pu at B18, 202.7 kW losses reproduced); 90 simulation, 45 interface and 15 browser tests.
- **Next:** valve actions in the planner, larger and real-topology networks, uncertainty in demand and repair times, and utility partnership for validation.

Research prototype — synthetic/benchmark infrastructure data. The electrical model is the IEEE 33-bus test feeder; the water network and district are synthetic and only Dubai-inspired. No private utility data is used; "Simulation Verified" is not a field certification.

Built by Team Kanban

1. M. E. Baran, F. F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," *IEEE Trans. Power Delivery*, 1989.
2. L. Thurner et al., "pandapower — an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Trans. Power Systems*, 2018.
3. K. A. Klise et al., "A software framework for assessing the resilience of drinking water systems to disasters with an example earthquake case study," *Environ. Model. Softw.*, 2017 (WNTR).
4. L. Rossman et al., *EPANET 2.2 User Manual*, U.S. EPA, 2020.

Live demo
encirra.kanbanstudios.ae

Source code
github.com/4waiz/Encirra